

Final Report

Auto-Ticket Classification using Flow Designer

Comprehensive Project Report

1. Abstract

This project implements an automatic ticket classification system for a school IT helpdesk using ServiceNow Flow Designer. Students submit support requests through a custom School IT Ticket form, and a Flow Designer workflow automatically classifies each ticket by detecting keywords in the short description. The solution eliminates manual category selection, reduces triage workload and provides immediate email acknowledgment to the requester.

2. Introduction

School IT helpdesk teams handle a variety of student requests ranging from Wi-Fi issues to hardware faults and password resets. Without automation, technicians must manually review and categorize every incoming ticket, a repetitive process that delays resolution. This project addresses that challenge by implementing a keyword-based classification flow within ServiceNow.

3. Objectives

- Simplify ticket submission by removing manual category selection.
- Automate classification using keyword detection in Flow Designer.
- Provide immediate email acknowledgment to students.
- Deliver pre-classified tickets to technicians for faster resolution.
- Track all customizations through ServiceNow Update Sets.

4. Literature Survey

#	Topic	Key Insight
1	ServiceNow Flow Designer	Event-driven, no-code automation for record-based workflows.
2	IT Service Management (ITSM)	Structured ticket handling improves response time and accuracy.
3	Keyword-Based Classification	Rule-based systems are effective for well-defined issue domains.
4	Email Notification Systems	Immediate acknowledgment builds trust and transparency.

5. Methodology

The project follows an Agile methodology with 5 sprints, each mapped to a project milestone:

Sprint	Milestone	Key Deliverables
--------	-----------	------------------

1	Requirement Analysis	Problem definition, Update Set creation
2	Backend Development	Custom table, fields, dependent choices
3	Flow Designer Automation	Classification flow, email notification
4	Testing & Validation	Functional testing, UAT
5	Deployment & Packaging	Update Set completion, XML export

6. Implementation

6.1 Custom Data Model

A custom School IT Ticket table was created with fields: Number (auto-generated), Caller (reference), Short Description (string), Category (choice), Subcategory (dependent choice), State, and Assignment Group.

<

≡

Dictionary Entry

New record

...

Submit

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the field **Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* Table

School IT Ticket [u_school_it_tick... ▼

* Type

String

Q

i

* Column label

Short description

* Column name

u_short_description

* Max length

160

Application

Global

i

Active

☒

Function field

☐

Mandatory

☐

Display

☐

Read only option

-- None -- ▼

Choice List Specification

Default Value

The **Default value** specifies what value the field has when first displayed.

Default value

Submit

Figure 1: Short Description field configuration.

Final Report | Page 4/12

servicenow

AllFavorites

Dictionary Entry - Caller

Search

⌵

⌕

⋮

⌵

<

≡

Dictionary Entry
Caller

🔗

⌵

⌵

⋮

Delete Column

Update

⬆

⬇

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the field **Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* TableSchool IT Ticket [u_school_it_tick...⌵

ApplicationGlobalⓘ

* TypeReference🔍ⓘ

Active☒

* Column labelCaller

Function field☐

* Column nameu_caller

Mandatory☐

Display☐

Read only option-- None --⌵

Reference Specification

Choice List Specification

Default Value

The **Default value** specifies what value the field has when first displayed.

Default value

Delete Column

Update

Figure 2: Caller reference field configuration.

servicenow

AllFavorites

Dictionary Entry - New Record

Search

⌵

⌕

⋮

⌵

<

≡

Dictionary Entry
New record

🔗

⌵

⌵

⋮

Submit

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the field **Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* TableSchool IT Ticket [u_school_it_tick...⌵

ApplicationGlobalⓘ

* TypeChoice🔍ⓘ

Active☒

* Column labelState

Function field☐

* Column nameu_state

Mandatory☐

Display☐

Max length

Read only option-- None --⌵

Choice List Specification

Default Value

The **Default value** specifies what value the field has when first displayed.

Default value

Submit

Figure 3: Category field configuration.

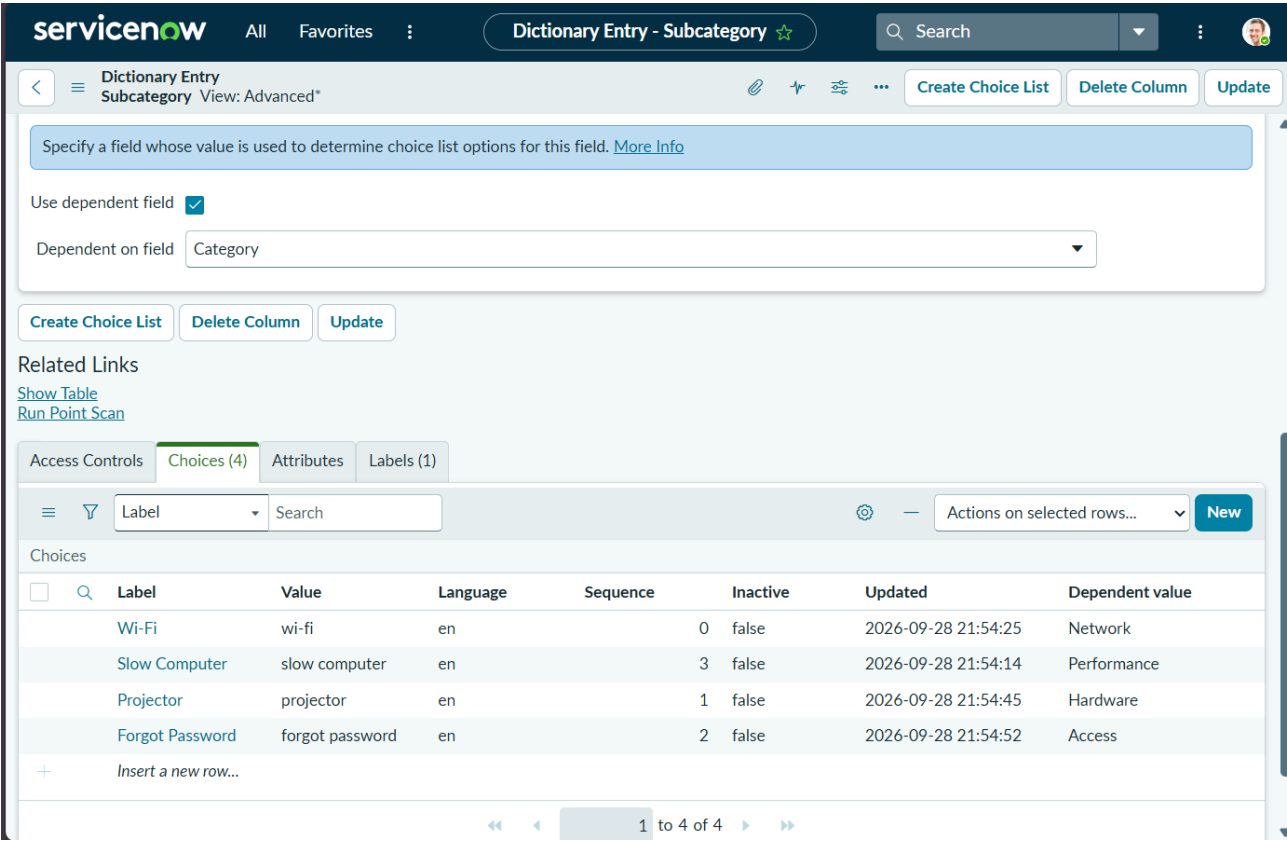


Figure 4: Subcategory configured as dependent on Category.

6.2 Flow Designer Workflow

The Auto Classify School IT Tickets flow triggers on Record Created. It evaluates the Short Description against keyword conditions and sets the appropriate Category and Subcategory values. An email action sends an acknowledgment to the Caller.

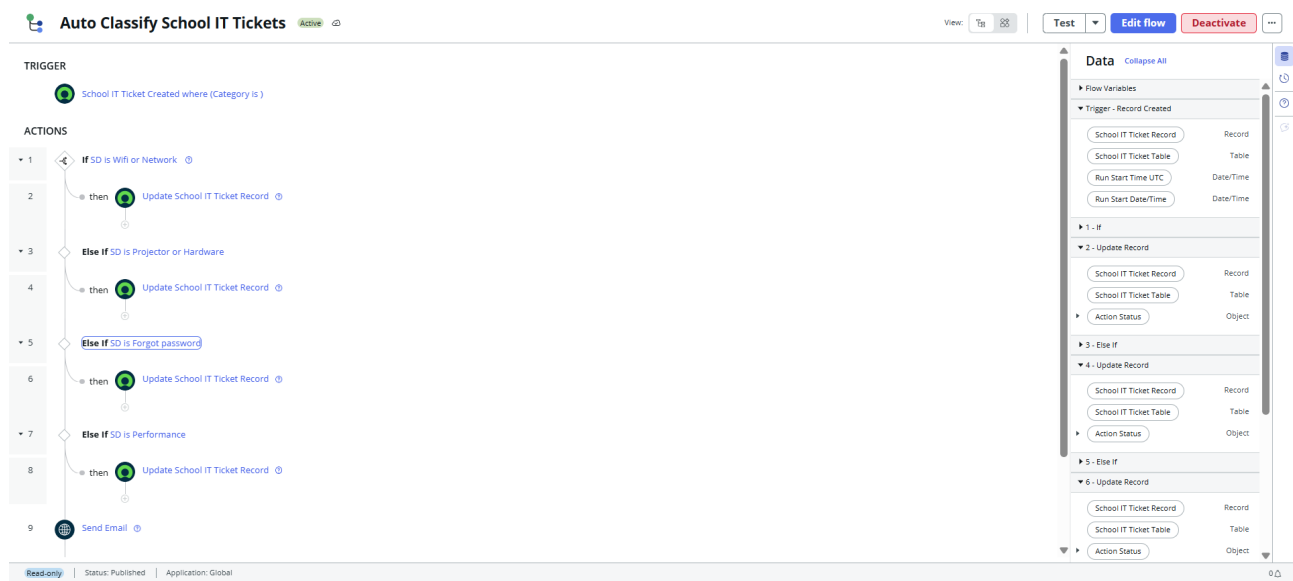


Figure 5: Published Auto Classify School IT Tickets flow.

6.3 Classification Rules

Keywords	Category	Subcategory
Wi-Fi, Network, Connectivity	Network	Wi-Fi / Connectivity
Projector, Hardware, Display	Hardware	Projector / Display
Password, Login, Access	Access	Password / Login
Slow, Performance, Lag	Performance	System Performance

6.4 Security & Change Tracking

ACLs and roles control access to the School IT Ticket table. All customizations (table definitions, field configurations, choice values and flow logic) are captured in the Project Update Set and exportable as XML for version control and deployment.

7. Results

All functional test cases and UAT scenarios passed with 100% acceptance rate.

Test Area	Cases	Passed	Failed
Keyword Classification	4	4	0
Email Notification	1	1	0
No-Match Handling	1	1	0

UAT Scenarios	5	5	0
---------------	---	---	---

7.1 Output Screenshots

servicenow

AllFavorites

School IT Ticket - TKT0001012

Search

<

≡

School IT Ticket
TKT0001012

UpdateDelete

↑↓

NumberTKT0001012

CallerSystem Administrator

Short descriptionWi-Fi not working in library

Description

CategoryNetwork

Assigned Group

SubcategoryWi-Fi

Assigned to

State-- None --

UpdateDelete

Figure 6: School IT Ticket showing automatic Category and Subcategory classification.

servicenow

AllFavorites

School IT Ticket - TKT0001013

Search

<

≡

School IT Ticket
TKT0001013

Update

Delete

Number

TKT0001013

Caller

System Administrator

Short description

Projector not turning on

Description

Category

Hardware

Assigned Group

Subcategory

Projector

Assigned to

State

-- None --

Update

Delete

Figure 7: Projector ticket with Hardware category auto-assigned.

servicenow

AllFavoritesHistoryWorkspaces

School IT Tickets

Search

≡

▼

🗨

School IT Tickets

Number

Search

🌐

🔍

📄

🕒

🔔

New

All

☐	🔍	Number	Assigned Group	Assigned to	Caller	Category	Description	Short description	State	Subcategory
		TKT0001010	(empty)	(empty)	(empty)			Wi-Fi not working in library		
		TKT0001013	(empty)	(empty)	System Administrator	Hardware	Projector not turning on			
		TKT0001011	(empty)	(empty)	System Administrator			Wi-Fi not working in library		
		TKT0001012	(empty)	(empty)	System Administrator	Network		Wi-Fi not working in library		

<<

<

1

>

>>

to 4 of 4

Figure 8: School IT Tickets list showing all created records.

servicenow

AllFavoritesHistory

Emails

Search

Actions on selected rows...New

All > Created on Today

Created	Recipients	Subject	Type	Notification type	User ID
2026-09-28 01:00:07	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-09-28 23:24:18	admin@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)
2026-09-28 10:01:15	admin@example.com	[ITSM Analytics] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-09-28 02:01:46	admin@example.com	[PA VSC] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-09-28 01:00:06	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-09-28 00:00:06	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-09-28 01:00:06	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)

1 to 7 of 7

Figure 9: ServiceNow Email log showing the acknowledgment notification.

Preview Email

Hello, your IT support ticket has been received and automatically categorized. Our support team will review it shortly.

Ref:MSG0000242_cUwO6n3sRXxEiaoCHSYx

Close

Figure 10: Acknowledgment email preview.

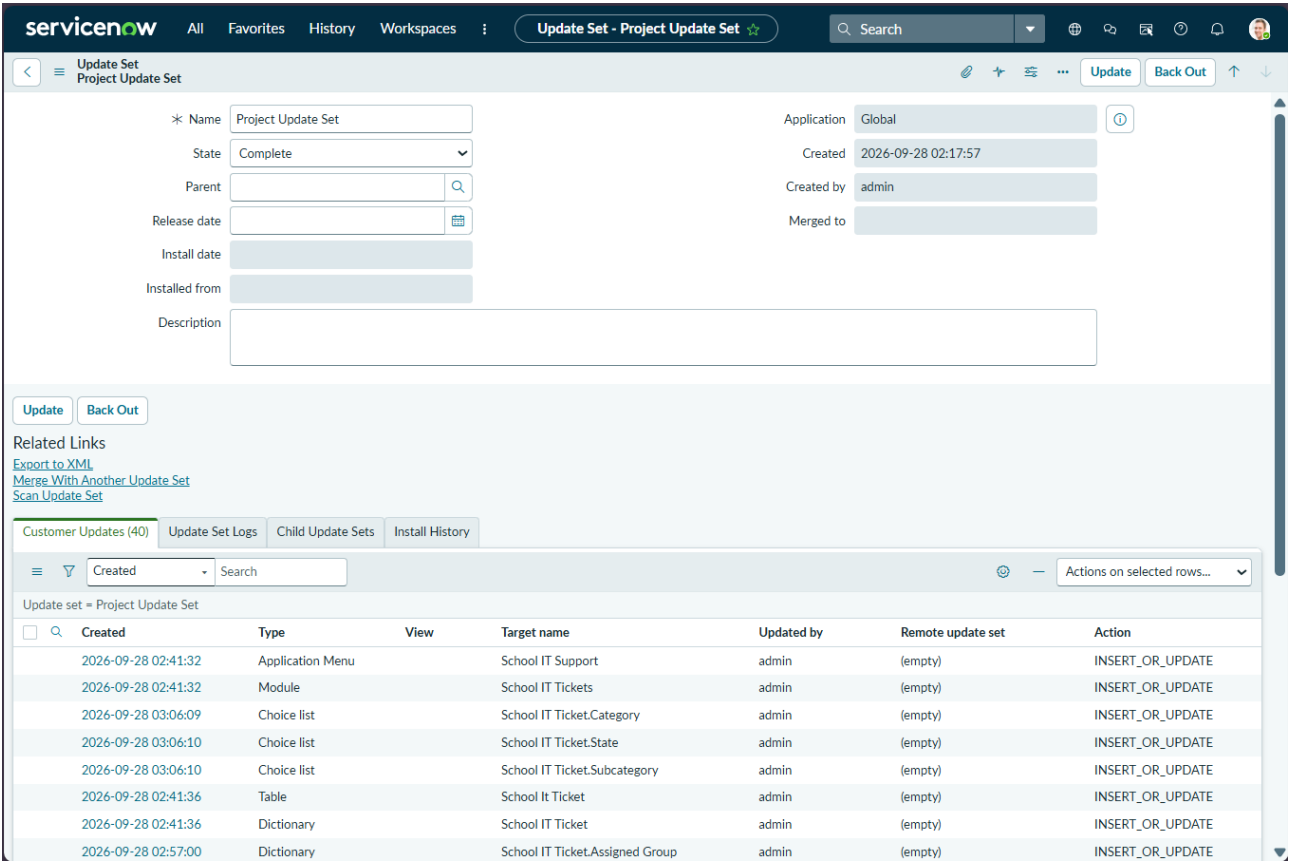


Figure 11: Completed Project Update Set with customer updates and XML export option.

8. Advantages & Disadvantages

Advantages

- Simple requester experience with basic input fields.
- Automatic classification for supported issue keywords.
- Consistent Category/Subcategory mapping.
- Immediate requester acknowledgment via email.
- Controlled change tracking through Update Sets.

Disadvantages

- Keyword rules depend on how the problem is described.
- Unsupported or vague descriptions require manual review.
- Current prototype is rule-based, not model-based.
- No automatic technician assignment implemented.

9. Conclusion

The Auto-Ticket Classification using Flow Designer project demonstrates a complete ServiceNow workflow for simplifying school IT helpdesk ticket processing. The implementation combines a custom data model, dependent choices, event-driven classification, email acknowledgment, access control, and Update Set based deployment tracking. All test cases passed, validating the solution's effectiveness.

10. Future Scope

- Expand keyword coverage and add fallback handling for ambiguous requests.
- Introduce automatic assignment to support groups.
- Add status-change and resolution notifications.
- Build dashboards for ticket volume, classification, and response trends.
- Evaluate AI-assisted classification as a future enhancement.

11. Appendix

Source Code: No separate application code - implemented via ServiceNow configuration.

Dataset: Not applicable - rule-based classifier, no ML dataset required.

GitHub: <https://github.com/Akash717656/Auto-Ticket-Classification-using-Flow-Designer>